# CS6848 — Principles of Programming Languages

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| Tier | B |
| Phase (order) | 6 |
| Do this first | CS2200, CS3300 |
| Estimated effort | 10-12 weeks |
| Unlocks | CS6225, CS6843 |

## Roadmap

1. Lambda calculus, evaluation strategies.
2. Type systems, polymorphism, subtyping.
3. Semantics (operational/denotational); write interpreters.

## Video courses (all links checked on YouTube on 2026-10-22)

### 🎓 Main course: follow it start to end

- Computer Sc - Principles of Programming Languages (40 videos). NPTEL *Principles of Programming Languages* (nptelhrd).

### 🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- MIT 6.001 Structure and Interpretation, 1986 (40 videos). MIT 6.001 *SICP* (Abelson & Sussman, 1986): the course that teaches what a programming language *is*.
- Introduction To Haskell Programming (29 videos). NPTEL-NOC IITM *Introduction to Haskell Programming* (Madhavan Mukund/S.P. Suresh).
- SML by Dan Grossman (19 videos). Dan Grossman *Programming Languages* (SML part, re-upload of the Coursera course): types, closures, pattern matching.

## Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

| Week | Lectures | What to do |
|---|---|---|
| 1 | 1–5 | Syntax, grammars, ambiguity, the PL0 language. |
| 2 | 6–9 | Semantics, syntactic classes, transition systems (operational semantics). |
| 3 | 10–19 | Binding, environments, declarations, commands, stores, blocks. Write an interpreter for a small imperative language in Python/OCaml. |
| 4 | 20–24 | Data, structured data, sequences, control, non-determinism. |

| Week | Lectures | What to do |
|---|---|---|
| 5 | 25–31 | Lambda calculus: data as functions, fixpoints, normal forms. Implement a lambda-calculus evaluator. |
| 6 | 32–35 | Monomorphism, polymorphism, type checking, contexts. Add type checking to your interpreter. |
| 7 | 36–40 | Abstractions, procedures, parameter passing. Then Dan Grossman's SML lectures (alt) for hands-on functional programming. |

▼ **All lectures in *Computer Sc - Principles of Programming Languages* (40 videos)**

1. Lecture - 1 Introduction to programming languages
2. Lecture - 2 Syntax
3. Lecture - 3 Grammars
4. Lecture - 4 Ambiguity
5. Lecture - 5 PLO:Syntax
6. Lecture - 6 Semantics
7. Lecture -7 Syntatic Classes
8. Lecture -8 Transition Systems
9. Lecture - 9 PL0 : Expressions
10. Lecture - 10 Binding
11. Lecture -11 Environments
12. Lecture -12 Declarations
13. Lecture -13 Commands
14. Lecture -14 Stores
15. Lecture -15 Summary
16. Lecture -16 Declarations and Commands
17. Lecture -17 Blocks
18. Lecture -18 Qualification
19. Lecture -19 Pragmatics
20. Lecture -20 Data
21. Lecture -21 Structured Data
22. Lecture - 22 Sequences
23. Lecture - 23 Control
24. Lecture - 24 Non- Determinacy
25. Lecture - 25 Programming Languages
26. Lecture - 26 Programming Languages
27. Lecture - 27 Programming Languages
28. Lecture - 28 Data as Functions
29. Lecture - 29 Data and Fixpoints
30. Lecture - 30 Normal Forms
31. Lecture - 31 Programming Languages

32. Lecture - 32 Monomorphism
33. Lecture - 33 Polymorphism
34. Lecture - 34 Type Checking
35. Lecture - 35 Contexts
36. Lecture - 36 Abstracts
37. Lecture - 37 Procedures
38. Lecture - 38 Meanings
39. Lecture - 39 Parameters
40. Lecture - 40 The Future

## Books, notes and websites

- Book: *Programming Languages: Application and Interpretation* (Krishnamurthi, free)
- Book: *Structure and Interpretation of Computer Programs* (SICP, free)
- ✅ NPTEL *Introduction to Haskell Programming* (S.P. Suresh, CMI) — onlinecourses.nptel.ac.in/noc21_cs77/preview
- Dan Grossman's *Programming Languages* (Coursera)

## Practice / proof of understanding

Write interpreters for lambda calculus and a typed mini-language.

## What your IITM course sheet says

### From the PDF: CS6848

*The course sheet has no syllabus text for this course; the plan below is based on the standard content of this subject.*